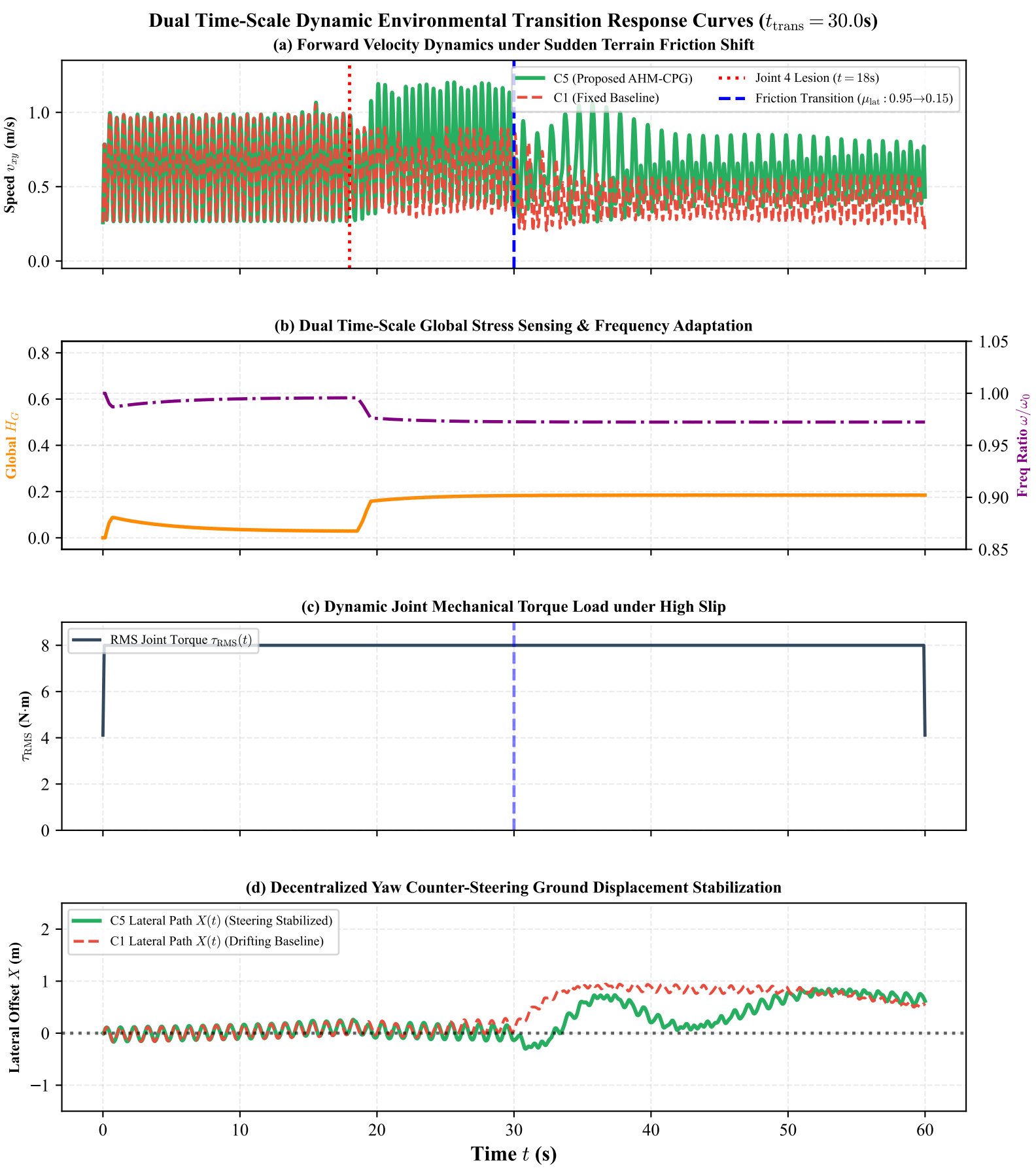

Dual Time-Scale Dynamic Environmental Transition Response Curves ($t_{\text{trans}} = 30.0$s)
(a) Forward Velocity Dynamics under Sudden Terrain Friction Shift
Speed v_{xy} (m/s)
C5 (Proposed AHM-CPG)
C1 (Fixed Baseline)
Joint 4 Lesion ($t = 18$s)
Friction Transition (μ_{lat} : 0.95→0.15)
(b) Dual Time-Scale Global Stress Sensing & Frequency Adaptation
Global H_G
Freq Ratio ω/ω_0
(c) Dynamic Joint Mechanical Torque Load under High Slip
RMS Joint Torque $\tau_{\text{RMS}}(t)$
τ_{RMS} (N·m)
(d) Decentralized Yaw Counter-Steering Ground Displacement Stabilization
C5 Lateral Path $X(t)$ (Steering Stabilized)
C1 Lateral Path $X(t)$ (Drifting Baseline)
Lateral Offset X (m)
Time t (s)